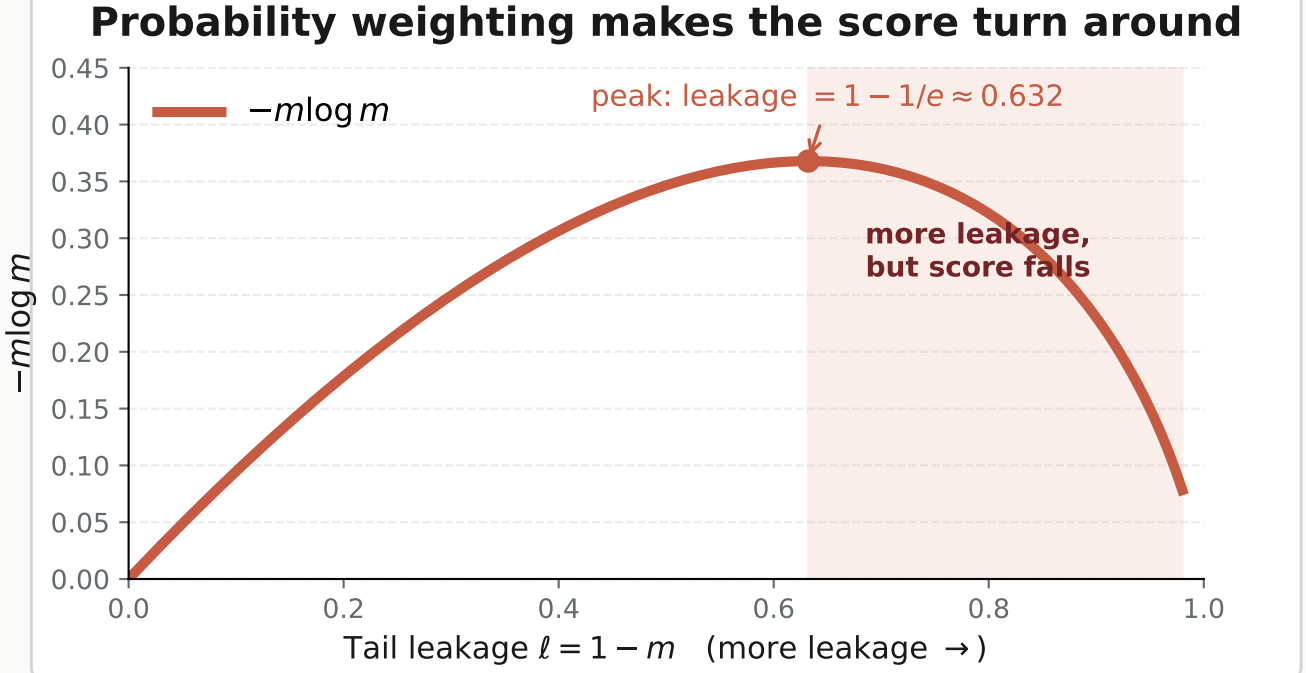
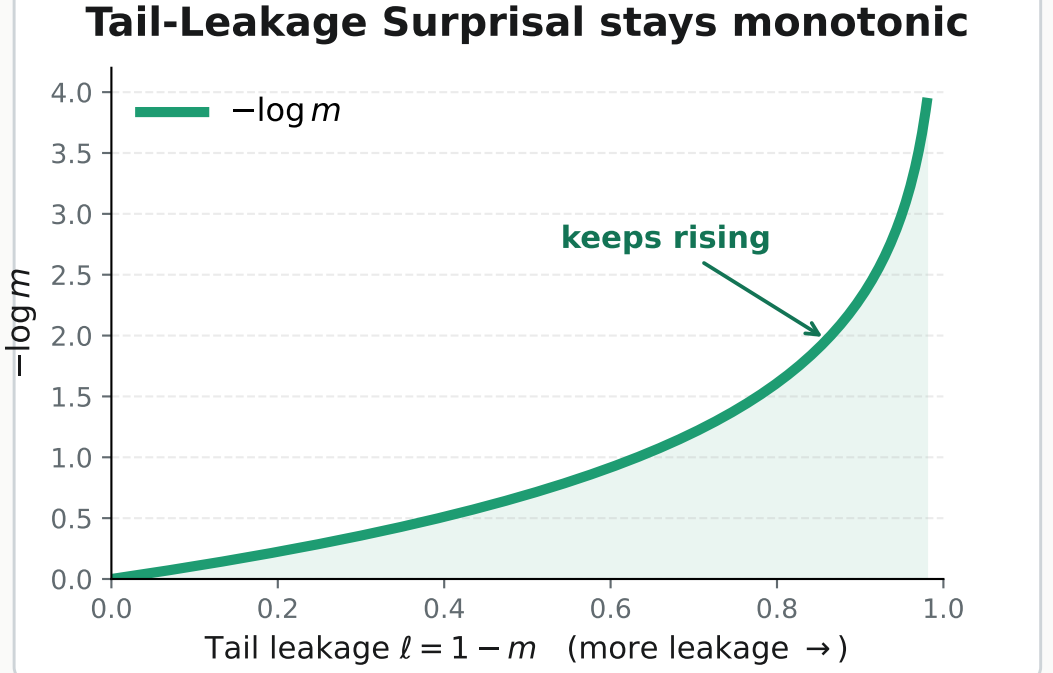
Why Not Use $-m\log m$ as the Final Risk Score?

A leakage score should never decrease when more probability escapes top-5.





$$-m\log m = m \times (-\log m)$$

The factor m suppresses the score when retained mass is already very small.

Takeaway: decomposition finds the variable m; $-\log m$ gives one-way leakage severity.

Design criterion: global monotonicity—not only fit within the observed m range.